

INTERNATIONAL GCSE CHEMISTRY

9202/2

Paper 2

Mark scheme

June 2025

Version: 0.1 Pre-Standardisation



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

Copyright information

OxfordAQA retains the copyright on all its publications. However, registered schools/colleges for OxfordAQA are permitted to copy material from this booklet for their own internal use, with the following important exception: OxfordAQA cannot give permission to schools/colleges to photocopy any material that is acknowledged to a third party even for internal use within the centre.

Copyright © 2025 OxfordAQA International Examinations and its licensors. All rights reserved.

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	Letter Sub-atomic particle A B C Electron Neutron Proton	allow 1 mark for 1 or 2 correct do not accept more than one line from a box on the left	2	AO1 3.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	group 3		1	AO2 3.1.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	a substance that is made of only one sort of atom		1	AO1 3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	isotopes		1	AO1 3.1.2g

Question	Answers			Extra information	Mark	AO/ Spec. Ref.
01.5	Sub-atomic particle	$^{12}_6\text{C}$	$^{14}_6\text{C}$			AO2 3.1.2e,f,g,h
	Electron	6	6		1	
	Neutron	6	8		1	
	Proton	6	6		1	

Total Question 1	8
------------------	---

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	Variable Dependent Independent Description of variable Change in mass of hydrogel Initial mass of hydrogel Time hydrogel left in water Type of hydrogel Volume of distilled water	do not accept more than one line from a box on the left	1 1	AO4 3.10.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	measured change in mass would be lower		1	AO4 3.10.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	(mass of water absorbed = $66.3 - 1.3 =$ 65 (g)) (ratio of the mass of water the hydrogel absorbed to the initial mass of the hydrogel =) $\frac{65}{1.3}$ $= 50 : 1$	allow correct use of an incorrectly determined mass of water absorbed	1 1 1	AO2 3.10.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	packaging		1	AO1 3.10.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	(thermosetting polymers) consist of polymer chains with cross links		1	AO1 3.10.2c
	(so thermosetting polymers) do not melt when heated		1	

Total Question 2		9
-------------------------	--	----------

Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	propane		1	AO1 3.10.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	C_nH_{2n+2}		1	AO1 3.10.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	all the carbon-carbon bonds are single covalent bonds		1	AO1 3.10.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	98 °C	allow a value in the range 93 °C to 103 °C	1	AO3 3.10.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	increases		1	AO3 3.10.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.6	(alkane X) has more carbon atoms in a molecule	allow (alkane X) is a larger molecule (than the alkanes in Figure 3)	1	AO2 × 2 AO3 × 1 3.10.1.1c
	(so alkane X) has a higher boiling point		1	
	(therefore X) condenses where the column has a higher temperature	allow (alkane X) collects where the column has a higher temperature	1	

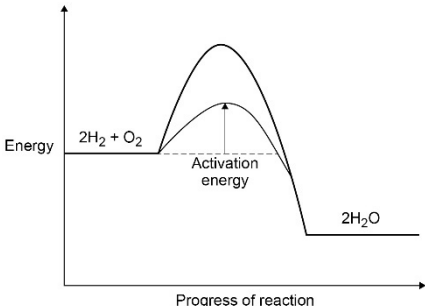
Total Question 3		8
-------------------------	--	----------

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	hydrogen fuel cells have a greater range (before refuelling in km)		1	AO3 3.9.3b
	(so) do not have to be refuelled as often		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	a hydrogen combustion engine travels further before replacement		1	AO3 3.9.3b 3.10.1.2d
	(so) will not be replaced as often		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	hydrogen is difficult to store		1	AO1 3.9.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	reaction line that starts on reactants and finishes on products, with a peak that is lower than the original activation energy labelled from reactant energy to top of drawn curve	an answer of  scores 2 marks	1 1	AO2 3.9.2d,h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	the energy of the products is lower than the energy of the reactants		1	AO2 3.9.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	acid rain		1	AO1 3.10.1.2d

Total Question 4		9
-------------------------	--	----------

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	giant structure		1	AO1 3.2.1i
	of atoms arranged in a regular pattern	allow atoms are arranged in layers	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	contains delocalised electrons		1	AO1 3.2.2g 3.3.1a
	(which) can move through metal / aluminium		1	

Question	Answers	Mark	AO/ Spec. Ref.
05.3	Level 2: Links are made and a judgement based on evidence provided.	3–4	AO3 3.3.1a,c
	Level 1: Relevant points are made. They are not logically linked.	1–2	
	No relevant content	0	
	Indicative content: - aluminium has a lower density - so the cables are lighter - aluminium has a lower cost per kilogram - so is cheaper to use - copper has a higher electrical conductivity - so is a better conductor - both have high melting points - so does not affect the use - reasoned judgement		

Total Question 5		8
-------------------------	--	----------

Question 6

Question	Answers	Mark	AO/ Spec. Ref.
06.1	Level 3: The design / plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.	5–6	AO4 3.9.2a
	Level 2: The design / plan would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The design / plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content: • measure known volume or mass of water • using measuring cylinder or mass balance • (into a) metal can or glass beaker • placed above spirit burner • measure initial temperature of water • with a thermometer (placed in the water) • measure the initial mass of the spirit burner • light spirit burner and heat up water • for a fixed period of time • or • until temperature rises a fixed amount • or • until mass of spirit burner drops 1 g • extinguish flame • measure final temperature • measure the final mass of spirit burner • calculate energy change (using $Q = mc\Delta T$) • divide energy change by mass of hydrocarbon burnt		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	as the size of the hydrocarbon molecule increases the energy released when burnt decreases		1	AO3 3.10.1.2c,e
	except for $C_{20}H_{42}$ which is anomalous		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	incomplete combustion	allow insufficient oxygen	1	AO1 3.10.1.2d

Total Question 6		9
-------------------------	--	----------

Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	the plants are burned		1	AO1 3.3.1.1f
	to produce ash (containing copper compounds)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	limit the environmental impact of traditional mining		1	AO1 3.3.1.1f
	(copper can be) extracted from low grade ores		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	(number of plants =) 4650 <u>0.775</u>		1	AO2 3.3.1.1f
	= 6000		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	a plant only absorbs a small mass of copper		1	AO2 3.3.1.1f
	plants take a long time to grow		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	(recycling) uses less energy		1	AO2 × 1
	(so) fewer fossil fuels are burnt		1	AO3 × 2 3.3.1.1i
	(so) less carbon dioxide is released		1	3.10.1.2g

Total Question 7		11
-------------------------	--	-----------

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	(hydrochloric) acid is an irritant	allow (hydrochloric) acid is corrosive	1	AO4 3.8.1a RPA4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	any one from: • bubbling / fizzing stopped • magnesium disappeared	allow the volume of gas in the measuring cylinder did not change	1	AO3 3.8.1a RPA4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	reduces escape of gas		1	AO4 3.8.1a RPA4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	all points plotted correctly	allow a tolerance of $\pm \frac{1}{2}$ a small square allow 1 mark for 4 or 5 points correctly plotted	2	AO2 $\times$ 2 AO3 $\times$ 1 3.8.1a RPA4
	line of best fit		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	90 (s)	allow a value in the range 88 (s) to 92 (s)	1	AO3 3.8.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.6	the rate of reaction decreases	allow (because) there are fewer (hydrochloric acid) particles per unit volume	1	AO2 3.8.1e
	(because) the concentration of hydrochloric acid decreases		1	
	(so) there is a lower frequency of collisions		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.7	(reproducible)			AO4 3.8.1a RPA4
	a different person obtains the same results		1	
	different equipment / techniques gives the same results		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.8	(volume conversion = $\frac{25}{1000}$ =) 0.025 dm^3	allow correct use of an incorrect / no conversion of volume allow correct use of an incorrectly determined value for moles	1	AO2 3.6.3b 3.6.4a
	(moles of H^+ in solution = 2.0×0.025 =) 0.05		1	
	(number of hydrogen ions =) $0.05 \times 6.02 \times 10^{23}$		1	
	= 3.01×10^{22}		1	

Total Question 8		16
------------------	--	----

Question 9

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.1	(strength)		1	AO3 3.3.1b
	(so) won't break (inside body)		1	
	(resistance to corrosion)			
	(so) will not react with anything (inside body)			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.2	($M_r \text{ TiO}_2 =$) $48 + (2 \times 16) = 80$		1	AO2 3.6.1d 3.6.2a 3.6.3a 3.6.4d
	$\left(\text{moles of TiO}_2 = \frac{720}{80} = \right) 9$	allow correct use of an incorrectly determined M_r of titanium dioxide	1	
	(moles of chlorine = $9 \times 2 =$) 18	allow correct use of incorrectly determined moles of titanium dioxide	1	
	(volume of chlorine =) 18×24	allow correct use of an incorrectly determined number of moles of chlorine	1	
	= 432 (dm ³)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.3	$\text{TiCl}_4 + 4 \text{ Na} \rightarrow \text{Ti} + 4 \text{ NaCl}$	allow for 1 mark $\text{TiCl}_4 + \text{Na} \rightarrow \text{Ti} + \text{NaCl}$ irrespective of balancing numbers	2	AO2 3.6.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.4	any one from: • violent reaction • produces a corrosive solution • produces hydrogen which is flammable	allow explosive	1	AO3 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.5	argon is unreactive oxygen (from air) would react with sodium / titanium		1 1	AO3 3.1.3c 3.7.1a

Total Question 9		12
-------------------------	--	-----------